

DATA SCIENCE B.S. DEGREE REQUIREMENTS

University: TechState University

Catalog Year: 2024–2025

Department: Statistics and Data Science

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1. OVERVIEW

The Bachelor of Science in Data Science is an interdisciplinary program jointly administered by the Department of Statistics and Data Science and the EECS Department. It requires a minimum of 124 credit hours.

2. MATHEMATICS AND STATISTICS CORE (28 credits)

- MATH 151: Calculus I (4 credits)
- MATH 152: Calculus II (4 credits) — prerequisite: MATH 151
- MATH 241: Linear Algebra (3 credits) — prerequisite: MATH 152
- MATH 251: Probability and Statistics for CS (3 credits) — prerequisite: MATH 152
- STAT 301: Statistical Inference (3 credits) — prerequisite: MATH 251
- STAT 302: Regression Analysis (3 credits) — prerequisite: STAT 301
- STAT 401: Bayesian Statistics (3 credits) — prerequisite: STAT 302 and MATH 241
- MATH 231: Discrete Mathematics (3 credits) — prerequisite: MATH 151

3. COMPUTING CORE (22 credits)

- CS 101: Introduction to Programming (3 credits)
- CS 102: Data Structures and Algorithms I (3 credits) — prerequisite: CS 101
- CS 201: Data Structures and Algorithms II (4 credits) — prerequisite: CS 102
- DS 210: Data Wrangling and Visualization (3 credits) — prerequisite: CS 101 and MATH 251
- DS 220: Big Data Technologies (3 credits) — prerequisite: CS 201 and DS 210
- DS 230: Database Systems for Data Science (3 credits) — prerequisite: CS 102 (Note: DS majors may substitute CS 260 for DS 230)
- DS 240: Machine Learning for Data Science (3 credits) — prerequisite: STAT 301 and CS 201 and MATH 241

4. DATA SCIENCE CORE (18 credits)

- DS 310: Data Ethics and Privacy (3 credits) — prerequisite: DS 210; no technical prerequisite
- DS 320: Applied Machine Learning (3 credits) — prerequisite: DS 240 and STAT 302
- DS 330: Time Series Analysis (3 credits) — prerequisite: STAT 302
- DS 340: Natural Language Processing for Data Science (3 credits) — prerequisite: DS 240

- DS 350: Capstone in Data Science (3 credits) — prerequisite: Senior standing, DS 320, and two additional DS 300-level courses
- DS 360: Data Science Consulting (3 credits) — prerequisite: DS 320; instructor consent required; limited to 15 students per semester

5. DOMAIN ELECTIVES (12 credits)

Students choose a domain emphasis from: Health & Medicine, Business Analytics, Environmental Science, or Social Science. Each domain has a set of 4 approved elective courses (3 credits each). Domain elective list is updated each academic year; students should consult the current year's supplemental elective sheet.

6. DOUBLE COUNTING WITH CS MAJOR

Students double-majoring in CS and Data Science may double-count up to 12 credits between the two programs, subject to advisor approval. CS 201, MATH 151, and MATH 152 may always be double-counted.

7. TRANSFER CREDIT POLICY

Transfer students must complete a minimum of 18 Data Science-prefix (DS) courses at TechState. CS transfer credits require evaluation by the EECS transfer credit committee. Statistics courses from accredited institutions transfer on a course-by-course basis with committee approval.

8. ACADEMIC REQUIREMENTS

Minimum cumulative GPA of 2.0 required. Students must earn at least C- in all required DS and STAT courses. DS 350 (Capstone) requires a minimum grade of C.